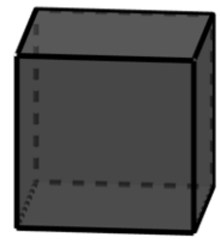
Grade 6 Accelerated Name_Answer Key_________
Unit 8 Date____________________
Lesson 1 Homework Period___________________

A $\frac{1}{3}$ inch unit cube is shown.

1. Determine the volume of the unit cube. Include the units for the cube's volume.

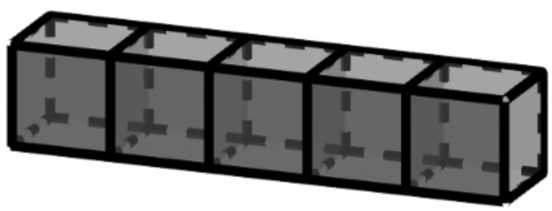


$\frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{27}$ in^3

The diagram shows five unit cubes whose dimensions are $\frac{1}{5}$ inches.

2. The volume of one cube is

__$\frac{1}{125}$__ cubic inches (in^3).

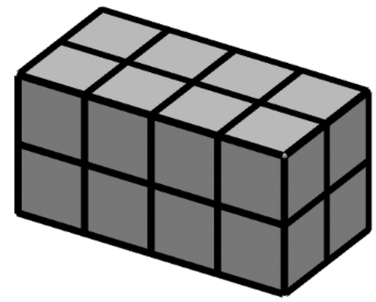


$\frac{1}{5} \cdot \frac{1}{5} \cdot \frac{1}{5} = \frac{1}{125}$ in^3

3. The volume of all five cubes is __$\frac{1}{25}$__ in^3.

$5 \cdot \frac{1}{125} = \frac{1}{25}$

A rectangular prism is shown.



4. How many unit cubes whose dimensions are $\frac{1}{4}$ inch fill up the rectangular prism?

16

5. Each unit cube whose dimensions are $\frac{1}{4}$ inch has a volume of $\frac{1}{64}$ cubic inches. Find the total volume, in cubic inches, represented in the diagram by multiplying the number of cubes that fill the rectangular prism by the volume of one of the cubes.

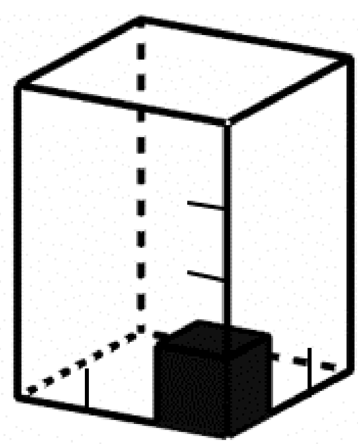
$16 \cdot \frac{1}{64} = \frac{1}{4}$ in^3

6. Find the total volume represented in the diagram, in cubic inches, using the formula, $V = lwh$.

$l = 1$ $\quad w = \frac{1}{2}$ $\quad h = \frac{1}{2}$

$V = 1 \cdot \frac{1}{2} \cdot \frac{1}{2}$ $\quad V = \frac{1}{4}$ in^3

A rectangular prism with one cube inside is shown.



7. How many cubes will it take to fill the rectangular prism?
3 cubes across, 3 cubes back, 4 cubes high
36 cubes

8. The rectangular prism has a length of $\frac{12}{5}$ centimeters (cm), a width of $\frac{12}{5}$ cm, and a height of $\frac{16}{5}$ cm as shown. Determine the dimensions of the fractional unit cube.

$\frac{16}{5} \div 4 = \frac{16}{5} \cdot \frac{1}{4} = \frac{4}{5}$ cm $\quad \frac{12}{5} \div 3 = \frac{12}{5} \cdot \frac{1}{3} = \frac{4}{5}$ cm

Each cube has dimensions of $\frac{4}{5}$ cm.

9. Determine the volume of the unit cube.

$\frac{4}{5} \cdot \frac{4}{5} \cdot \frac{4}{5} = \frac{64}{125}$ cm^3

10. Determine the volume of the rectangular prism.

$36 \cdot \frac{64}{125} = \frac{2{,}304}{125} = 18\frac{54}{125}$ cm^3

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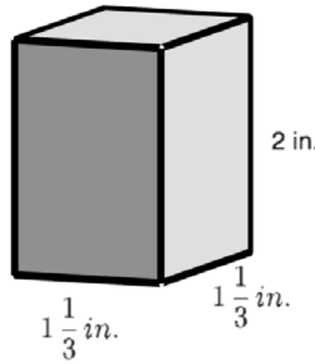
Lesson 2 Homework Period___________________

A rectangular prism is shown with units measured in inches.



In questions 1–4, find the volume using cubes with a side length of $\frac{2}{3}$ in.

1. Determine how many cubes will fit along the length, width, and height.

$1\frac{1}{3} \div \frac{2}{3} =$ ___2_____ cubes will fit along the length.

$\frac{4}{3} \div \frac{2}{3} = {}^{2}\frac{4}{\cancel{3}} \times \frac{\cancel{3}}{4}{}_{1} = 2$

$1\frac{1}{3} \div \frac{2}{3} =$ ___2_____ cubes will fit along the width.

$2 \div \frac{2}{3} =$ ___3_____ cubes will fit along the height.

$2 \div \frac{2}{3} = {}^{\cancel{2}} \times \frac{3}{\cancel{2}} = 3$

2. A total of ___12____ cubes will fit within the rectangular prism.

3. The volume of each cube is $\frac{8}{27}$ in^3. Multiply the number of cubes needed to fill the prism by the volume of each cube in order to find the volume of the prism. Include units.

$4\ \cancel{12} \cdot \frac{8}{\cancel{27}}{}_{9} = \frac{32}{9} = 3\frac{5}{9}$ in^3

4. Instead of using cubes, Kaia uses the formula $V = l \times w \times h$. Fill in the spaces below to complete her work. Include units.

$V =$ ___$1\frac{1}{3}$___ $\times$ ___$1\frac{1}{3}$___ $\times$ ___2___ $\frac{4}{3} \times \frac{4}{3} \times \frac{2}{1} = \frac{32}{9}$

$V =$ ___$3\frac{5}{9}$____ in.^{3}

A baker is packaging her cupcakes into single serve boxes for a special event. Each cube-shaped box has side lengths 3.5 inches (in.). The smaller boxes are then packed into a larger box with dimensions of 10.5 in. $\times 7$ in. $\times 7$ in.

5. How many single serve cupcakes can be packed into the larger box?

$10.5 \div 3.5 = 3$

$7 \div 3.5 = 2$ 12 boxes can be packed into the larger box.

$7 \div 3.5 = 2$

6. Use the number of cupcakes that can be packed in the larger box to determine the volume of the box. Include units.

$(3.5 \times 3.5 \times 3.5) \times 12 = 42.875 \times 12 = 514.5$ in.^{3}

Owen and Noah are calculating the volume of a rectangular prism with the dimensions of 2 cm $\times$ 2 cm $\times$ 1 cm using different size cubes. Noah uses cubes with side lengths lengths $\frac{1}{3}$ cm and Owen uses cubes with side lengths $\frac{1}{6}$ cm.

7. Who will need more cubes to fill the larger prism? Justify.

$2 \div \frac{1}{3} = 6$		$2 \div \frac{1}{6} = 12$	
$2 \div \frac{1}{3} = 6$	108	$2 \div \frac{1}{6} = 12$	864
$1 \div \frac{1}{3} = 3$	cubes	$1 \div \frac{1}{6} = 6$	cubes

Owen will need more cubes because his cubes are smaller.

8. What is the volume of the rectangular prism? Include units.

$2 \times 2 \times 1 = 4$ cm^3

9. If they had used a cube with side lengths $\frac{1}{4}$ cm instead, would they have found the same volume of the rectangular prism?

Yes because the dimensions of the prism didn't change.

10. An aquarium in the shape of a rectangular prism is being filled with water. The dimensions of the aquarium are 48.5 inches (in.) by 18.25 in. If the height of the water is 15.75 in., what is the volume of the water in cubic inches?

$48.5 \times 18.25 \times 15.75 = 13{,}940.71875$ or $13{,}940\frac{23}{32}$

Grade 6 Accelerated Name___Answer Key_______
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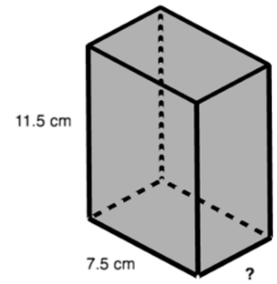
Lesson 3 Homework Period___________________

A rectangular prism is shown with measurements in centimeters (cm).

1. What is the area of the front face of this prism? Include units.



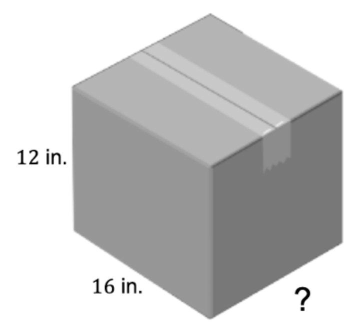
$11.5 \times 7.5 = 86.25$ cm^2

2. If the volume of the box is 345 cubic centimeters (cm^3), what is the value of the missing dimension of the box?

$\frac{345}{86.25} = 4$, 4 cm

Stefani is packing moving boxes. The boxes have a length of 16 inches (in.) and a height of 12 in., but she cannot recall the height of the box.

3. Determine the area of the front face of the box. Include units.



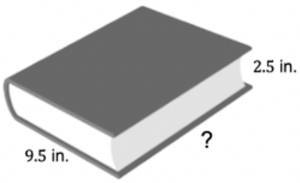
$12 \times 16 = 192$ in^2.

4. If the volume of the box is $2{,}304$ cubic inches (in^3), what is the width of the box? Include units.

$\frac{2304}{192} = 12$ in.

Jeremiah was trying to find the missing dimension of the journal shown. He made a mistake when solving for the width of the box, but he cannot find it. His work is shown.

Volume = cubic in.



Jeremiah's work	
$\begin{array}{r} 9.5 \\ \times \quad 2.5 \\ 455 \\ \underline{+\,1900} \\ 2355 \end{array}$ $\rightarrow 23.55$ in^2	$285 \div 23.55 = 12.1$ $w = 12.1$ in.

5. Circle the mistake.

6. What should the width of the box be? Include units.

$285 \div 23.75$ in. $= 12$ in.

7. Write a note to Jeremiah on how to correct his error(s).
Don't forget to carry the tens values when finding products; you should have carried the 2 when multiplying 5×5.

8. A box of crayons has a length of 1.5 inches (in.) and a height of 4 in. The box has a volume of 16.8 cubic inches (in^3). What is the width of the box? Include units.

$\frac{16.8}{(1.5 \times 4)} = \frac{16.8}{8} = 2.8$ in.

9. Josh bought a sandbox. He knows that the length is 3.5 feet (ft) and the width is 2.5 ft., but he can't remember the appropriate height. The volume needs to be 10.5 cubic feet (ft^3). What is the height of the sandbox? Include units.

$\frac{10.5}{(2.5 \times 3.5)} = \frac{10.5}{8.75} = 1.2$ ft.

10. Cara is measuring her suitcase before her trip. The case has a width of 8.75 inches (in.) and a height of 14 in. The volume of the case is $2{,}756.25$ cubic inches (in^3). Determine the length of the suitcase. Include units.

$\frac{2756.25}{(8.75 \times 14)} = \frac{2756.25}{122.5} = 22.5$ in.

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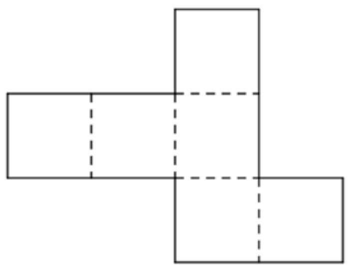
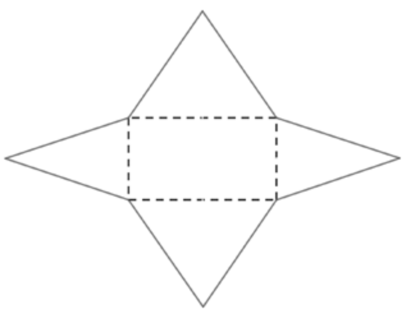
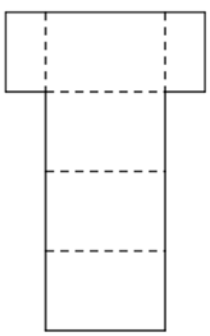
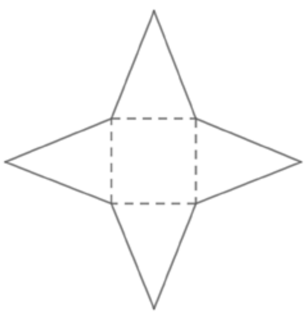
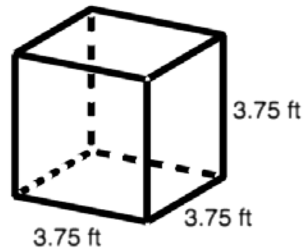
Lesson 4 Homework Period____________________

Complete the table to describe the three-dimensional figure formed for questions 1–4.

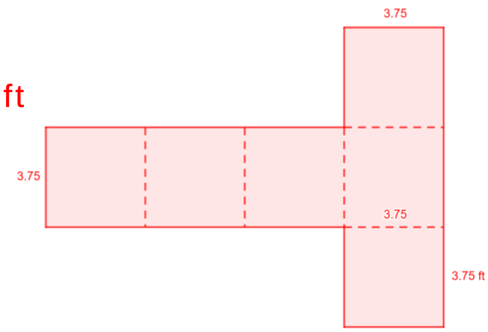
	Net	Three-Dimensional Figure Formed
1.		Cube
2.		Rectangular pyramid
3.		Rectangular prism
4.		Pyramid

A three-dimensional figure shown for questions 5–6.



5. Sketch the net or the individual faces of the three-dimensional figure.

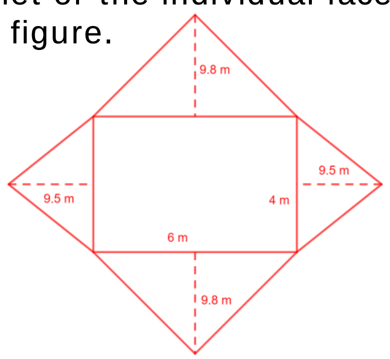
*all dimensions are 3.75 ft

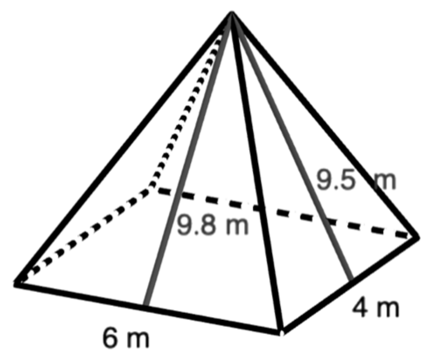


6. Label each side of the net with the appropriate measurement in feet (ft).

A three-dimensional figure is shown for questions 7–8.

7. Sketch the net or the individual faces of the three-dimensional figure.

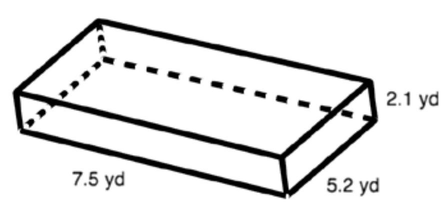


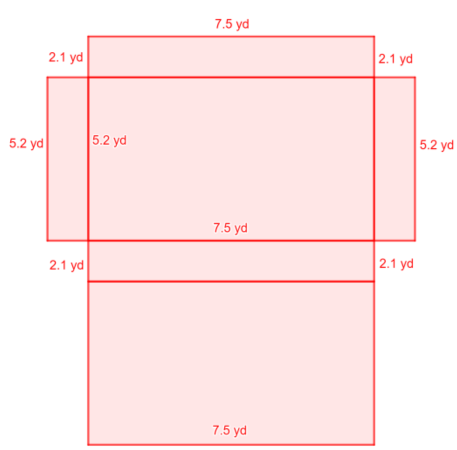


8. Label each side of the net with the appropriate measurement in meters (m).

A three-dimensional figure is shown for questions 9–10.

9. Sketch the net or the individual faces of the three-dimensional figure.





10. Label each side of the net with the appropriate measurement in yards (yd).

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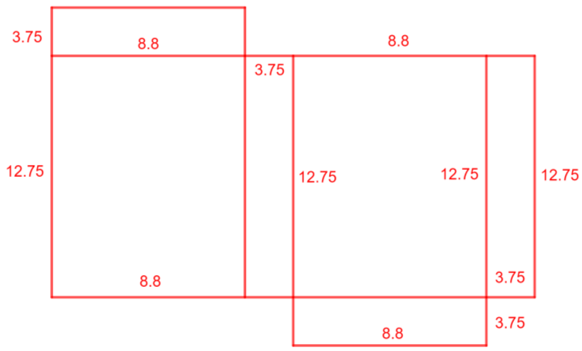
Lesson 5 Homework Period___________________

A box of cereal measures 8.8 inches (in.) long, 3.75 in. wide, and 12.75 in. tall.

1. Sketch the net of the cereal box.



2. Label the dimensions of each face on the net.

3. Find the area of each face. Include units.

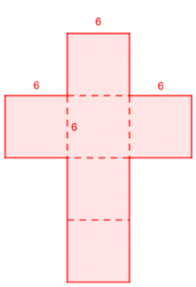
Front: $8.8 \times 12.75 = 112.2$ in^2
Back: $8.8 \times 12.75 = 112.2$ in^2
Side: $12.75 \times 3.75 = 47.8125$ in^2
Side: $12.75 \times 3.75 = 47.8125$ in^2
Top: $8.8 \times 3.75 = 33$ in^2
Bottom: $8.8 \times 3.75 = 33$ in^2

4. Add the area of the faces to find the surface area of the cereal box. Include units.

$2(112.2) + 2(47.8125) + 2(33) = 224.4 + 95.625 + 66$
$= 386.025$ in^2

Sharee is designing boxes for the rings she sells. The ring box has measurements 6 centimeters (cm) long, 6 cm wide, and 6 cm high.

5. Sketch and label the net of the ring box.
All sides are 6 cm.



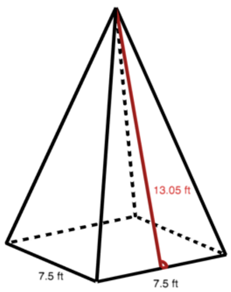
6. What is the surface area of the ring box? Include units.

Front/back: $2(6 \times 6) = 72$ cm^2
Side/Side: $2(6 \times 6) = 72$ cm^2
Top/Bottom: $2(6 \times 6) = 72$ cm^2

$72.3 = 216$ cm^2

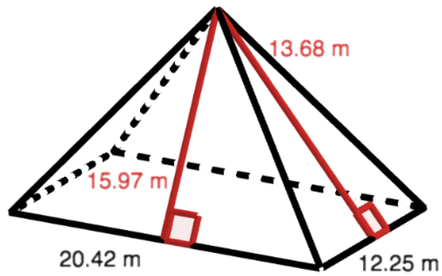
For questions 7–8, find the surface area each pyramid. Include units.

7.



$\square$: $7.5 \times 7.5 = 56.25$ ft^2
$\triangle$: $\frac{1}{2} \times 7.5 \times 13.05 = 48.9375$ ft^2
$56.25 + 4(48.9375)$
$= 252$ ft^2

8.



$\square$: $20.42 \times 12.25 = 250.145$ m^2
$\triangle_1$: $\frac{1}{2} \times 20.42 \times 15.97 = 163.0537$ m^2
$\triangle_2$: $\frac{1}{2} \times 12.25 \times 13.68 = 83.79$ m^2
$250.145 + 2(163.0537) + 2(83.79)$
$= 743.8324$ m^2

9. Calculate the surface area of a rectangular pyramid that has a square base with a side length of 14.5 centimeters (cm), which is the base of the triangle with a slant height of 14.2 cm.

$14.5 \times 14.5 = 210.25$ cm^2
$\frac{1}{2} \times 14.5 \times 14.2 = 102.95$ cm^2
$210.25 + 4(102.95) = 622.05$ cm^2

10. Calculate the surface area of a rectangular pyramid that has a rectangular base with a side length of 12 yards (yd), which is the base of a triangle with a slant height of 11.66 yd, and a side length of 15 yd, which is the base of a triangle with a slant height of 12.5 yd.

$12 \times 15 = 180$ yd^2
$\frac{1}{2} \times 12 \times 11.66 = 69.96$ yd^2
$\frac{1}{2} \times 15 \times 12.5 = 93.75$ yd^2
$180 + 2(69.96) + 2(93.75) = 507.42$ yd^2

Grade 6 Accelerated
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Lesson 6 Homework

Name ______Answer Key______
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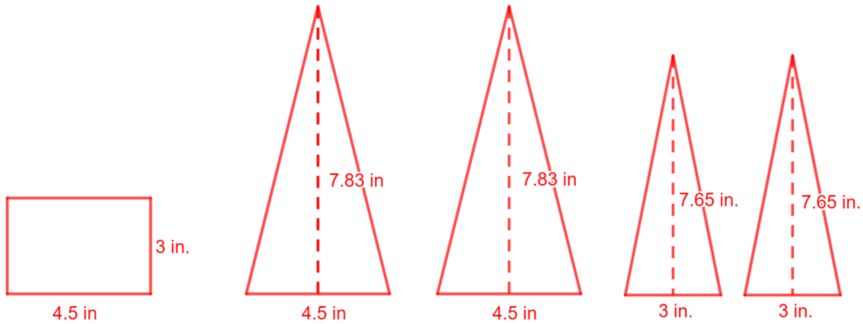
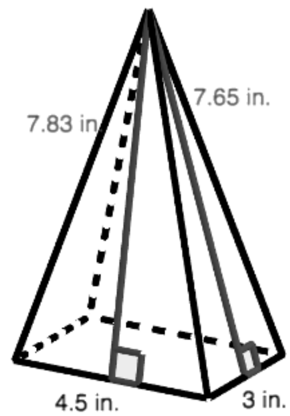
A rectangular pyramid is shown with measurements in inches (in.).

1. Draw a net or the individual faces of the figure. Label the measurements.

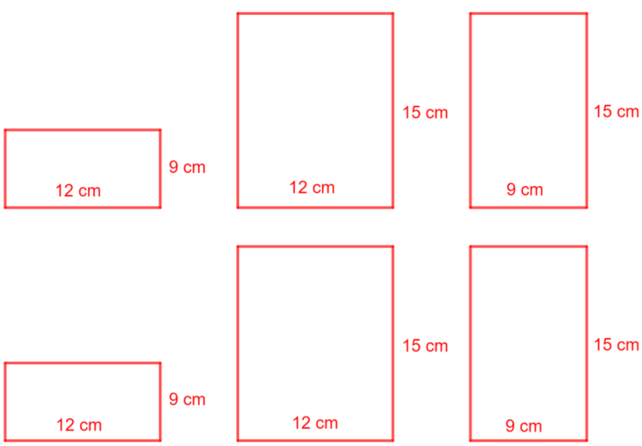


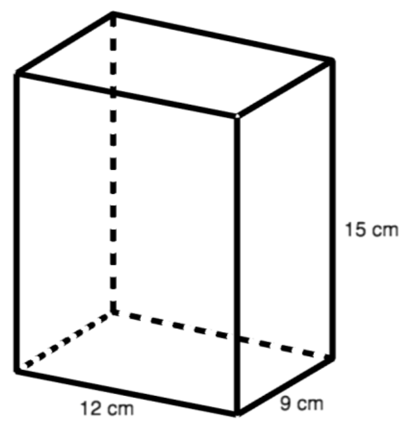


2. Find the surface area of the figure.
$135 + 2(0.5 \times 4.5 \times 7.83) + 2(0.5 \times 3 \times 7.65)$
$= 13.5 + 2(17.6175) + 2(11.475)$
$= 13.5 + 35.235 + 22.95$
$= 71.685$ in.^{2}

A rectangular prism is shown with measurements in centimeters (cm).

3. Draw a net or the individual faces of the figure. Label the measurements.





4. Find the surface area of the figure.
$2(108) + 2(180) + 2(135)$
$= 216 + 360 + 270$
$= 846$ cm^2

5. Find the volume of the figure.
$12 \times 9 \times 15 = 1{,}620 \text{ cm}^3$

Sanjay is designing a suitcase. He drew a model of his suitcase.

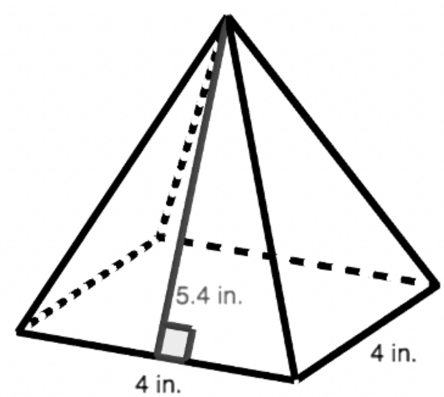
6. If the case has a volume of $7{,}098$ cubic inches (in^3), what is the length of the case?

$$\frac{7098}{(13 \times 28)} = \frac{7098}{364} = 19.5 \text{ in.}$$

7. How much material would Sanjay need to make a case with these dimensions, not taking into consideration any overlap of the faces?
$2(19.5 \times 28) = 2(546) = 1{,}092$
$2(19.5 \times 13) = 2(253.5) = 507$
$2(13 \times 28) = 2(364) = 728$
$1{,}092 + 507 + 728 = 2{,}327$ in^2

8. If the inside of the case has a length of 18 in., a width of 11.5 in., and a height of 24 in., how much smaller is the volume of the inside of the case than the volume of the entire case?
$7098 - (18 \times 11.5 \times 24) = 7098 - 4968 = 2{,}130$ in^3 smaller

Jocelyn makes and sells bookends that are in the shape of rectangular pyramids.



9. What is the surface area of one bookend?
$4 \times 4 = 16$ in^2
$0.5 \times 4 \times 5.4 = 10.8$ in^2
$16 + 4(10.8) = 59.2$ in^2

10. If Jocelyn paints only the triangular faces, what is the surface area of the area she wants to paint?
$4(10.8) = 43.2$ in.^{2}